From Wigner's Surmise to the Marchenko–Pastur Law: An Account of Random Matrix Theory, with an Application to NSE Equity Portfolios

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Abstract

This paper is an expository account of the random matrix theory I have studied, organized as the subject itself unfolded: what a random matrix is, the two inequivalent ways a matrix ensemble can be structureless—independent entries, or rotational invariance—and the proof, by a two-by-two trace computation, that the Gaussian ensembles (GOE, GUE) are exactly where the two properties meet; Wigner's surmise for the spacing of adjacent eigenvalues, including the calculation from which it takes its name; the Wishart ensemble, which is what a random matrix looks like when it arises as a sample covariance matrix; and the Marchenko–Pastur theorem, the spectral law of Wishart matrices, derived here in full using the Stieltjes transform and its inversion formula, the Sherman–Morrison rank-one update, Sylvester's determinant identity, and the concentration of quadratic forms—with a careful discussion of why the theorem lives in the limit where both matrix dimensions diverge at a fixed ratio $\gamma = N/T$. The final part puts the theory to work on data I assembled: daily returns of 548 National Stock Exchange of India equities over 1236 trading days. The eigenvalue bulk of the sample correlation matrix matches the Marchenko–Pastur density, its level spacings match Wigner's surmise (and decisively reject the Poisson alternative), and a covariance matrix cleaned by flattening the noise band delivers out-of-sample minimum-variance portfolios with 7.8% annualized volatility against 9.3% for Ledoit–Wolf shrinkage, 17.7% for equal weights, and 19.4% for the raw matrix. Nothing in the theory below is new; the aim is to show, end to end, how these classical results connect and what they can do when applied carefully.

Acknowledgements

This paper is a record of self-study, and its debts are to teachers I have never met. The *Linear Algebra* and *Abstract Linear Algebra* playlists of the YouTube channel **The Bright Side of Mathematics** built the foundations on which everything here stands. **Roland Speicher**'s lecture playlist on high-dimensional random matrices taught me how the subject thinks in the large-N limit. **Euler Bernhard**'s *Random Matrices: Theory and Practice* playlist walked me through the ensembles and their spectral laws with exactly the mix of rigor and intuition I have tried to reproduce in Sections 3–5. And **Terence Tao**'s writing on random matrix theory showed me what it looks like when deep mathematics is explained with total clarity. Any errors of understanding in these pages are, of course, entirely my own.

1 Introduction

This is not a research paper in the usual sense. It is a record of understanding: an attempt to lay out, in one connected narrative, a body of classical mathematics I have studied—random matrix theory, from Wigner's work in the 1950s to the Marchenko–Pastur theorem of 1967—and then to show that the understanding is real by applying it to a concrete problem: estimating the covariance matrix of Indian equities well enough to build portfolios that survive contact with unseen data.

The narrative has a natural arc, and the paper follows it. Section 2 explains what random matrix theory is and the physical question that created it. Section 3 maps the landscape of ensembles—matrices with *independent entries* on one side, *rotationally invariant* ensembles on the other, and the Gaussian ensembles (GOE, GUE, GSE) living precisely in their intersection, a fact proved by a short trace computation that was the first genuine proof I learned in this subject—and then presents Wigner's surmise for eigenvalue spacings, with the two-by-two calculation behind it. Section 4 explains why finance forces a different ensemble: covariance matrices are not symmetric matrices with independent entries but matrices of the form $X^\top X$, whose ensemble is Wishart's and whose spectral law is Marchenko–Pastur's. Section 5 is the heart of the paper: a complete derivation of the Marchenko–Pastur density from the Wishart setup, built from four tools—the Stieltjes transform and its inversion, the Sherman–Morrison update, Sylvester's determinant identity, and concentration of quadratic forms—each stated and proved or sketched before it is used, together with an explanation of why the limit is taken with both dimensions growing at fixed ratio. Section 6 then applies everything at once to five years of NSE data, testing the claim that the in-band eigenvalues are genuinely noise in three layers of increasing strictness: the Marchenko–Pastur density confirms they are *distributed* like noise, Wigner's surmise confirms they *interact* like noise, and an out-of-sample portfolio backtest—in which discarding the band's contents was free to backfire—confirms they *price* like noise, beating the raw matrix, Ledoit–Wolf shrinkage, and equal weights decisively. Section 7 reflects on what the exercise taught.

Everything in Sections 2–5 is classical material, drawn from the literature [1, 2, 4, 5, 8]; the data work in Section 6 is my own, in the tradition started for financial correlation matrices by Laloux et al. [6] and Plerou et al. [7].

2 What random matrix theory is

Random matrix theory begins with a confession of ignorance. In the 1950s, Eugene Wigner faced nuclear energy spectra too complicated to compute from first principles, and made a radical move: model the unknown system as a matrix with *random* entries, constrained only by basic symmetry, and ask what its eigenvalues look like *typically.* The gamble was that quantities like the spectrum's shape and the gaps between neighbouring levels don't depend on microscopic detail at all—that they are *universal*—so a maximally ignorant model predicts them correctly. The gamble won: level statistics across nuclei, atoms, and chaotic quantum systems of every kind match the random-matrix predictions, and the same universality later turned out to govern systems with no physics at all, from traffic flow to financial correlations.

The canonical model is the *Gaussian Orthogonal Ensemble* (GOE): a real symmetric $N \times N$ matrix H whose entries on and above the diagonal are independent Gaussians, $H_{ii} \sim \mathcal{N}(0, 1)$ and $H_{ij} \sim \mathcal{N}(0, \frac{1}{2})$ for $i < j$. (The variance convention makes the distribution invariant under every orthogonal change of basis $H \mapsto O^\top HO$—"no basis is special," the mathematical form of

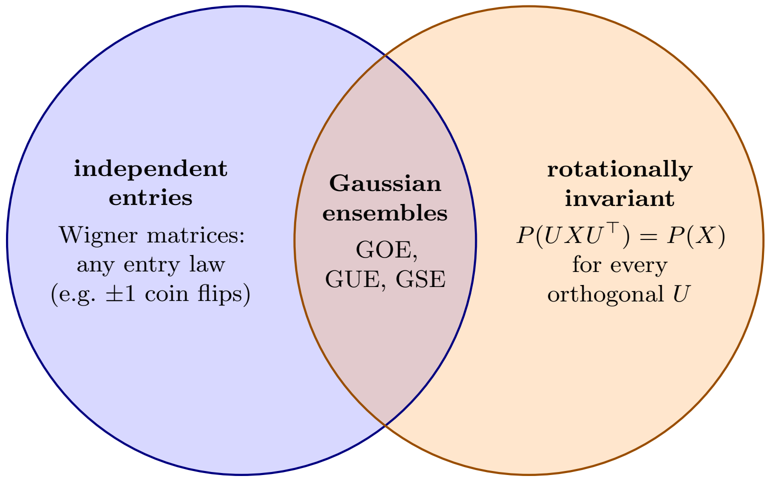

Figure 1: The landscape of random matrix ensembles. One can demand statistical structurelessness (independent entries) or geometric structurelessness (no preferred basis). The two demands are genuinely different—and the Gaussian ensembles are, essentially uniquely, where they meet.

maximal ignorance.) As N grows, what do the eigenvalues $\lambda_1, \ldots, \lambda_N$ of such a matrix do?

Two kinds of answer exist, and the distinction organizes everything that follows. *Global* statistics describe where the eigenvalues lie on average—the limiting histogram, or spectral density. *Local* statistics describe how neighbouring eigenvalues space themselves—whether they cluster, ignore each other, or repel. For the covariance matrices of finance, the global answer is the Marchenko–Pastur law derived in Section 5; the local answer, which universality makes common to essentially all of these ensembles, is Wigner's surmise. Before either, we should understand the ensembles themselves.

3 The Gaussian ensembles, and Wigner's surmise

3.1 Two ways to be structureless, and where they meet

A random matrix model must decide what "no structure" means, and there are two natural, inequivalent answers (Figure 1). The first is *statistical*: fill the matrix with **independent entries**, of whatever distribution—Gaussians, coin flips, fat-tailed returns. Matrices of this kind (symmetric, with independent entries above the diagonal) are called *Wigner matrices*. The second is *geometric*: demand that the ensemble have **no preferred basis**, i.e. that the probability density be *rotationally invariant*,

$$P(UXU^\top) = P(X) \qquad \text{for every orthogonal } U,$$

so that the model looks identical in every coordinate system—the mathematical form of Wigner's "maximal ignorance." These two demands are genuinely different. A symmetric matrix of independent ± 1 coin flips is a perfectly good Wigner matrix, but rotating it produces entries that are neither ± 1 nor independent; conversely, a typical invariant ensemble has correlated entries. Neither set contains the other.

The remarkable fact is that the intersection is essentially a single point: the *Gaussian* ensembles, and nothing else [2]. I found the direct verification more illuminating than the abstract statement, so here it is for the 2×2 real symmetric case. Take

$$X = \begin{pmatrix} x_1 & x_3 \\ x_3 & x_2 \end{pmatrix}, \qquad x_1, x_2 \sim \mathcal{N}(0,1), \quad x_3 \sim \mathcal{N}(0, \tfrac{1}{2}),$$

all independent. The joint density is the product of the three marginals,

$$P(X) = P(x_1)P(x_2)P(x_3) = \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{\sqrt{\pi}}\, e^{-\frac{1}{2}x_1^2 - \frac{1}{2}x_2^2 - x_3^2} = \frac{1}{2\pi\sqrt{\pi}}\, e^{-\frac{1}{2}\left(x_1^2 + x_2^2 + 2x_3^2\right)}.$$

Now compute X^2: its diagonal entries are $x_1^2 + x_3^2$ and $x_2^2 + x_3^2$, so

$$\operatorname{tr} X^2 = x_1^2 + x_2^2 + 2x_3^2,$$

exactly the combination in the exponent. The three independent marginals have conspired to produce a density that depends on the matrix only through a trace:

$$P(X) = \frac{1}{2\pi\sqrt{\pi}}\, e^{-\frac{1}{2}\operatorname{tr} X^2}. \tag{1}$$

Rotational invariance is now immediate: for any orthogonal U,

$$\operatorname{tr}\Big[(UXU^\top)^2\Big] = \operatorname{tr}\Big[UX^2U^\top\Big] = \operatorname{tr} X^2$$

by cyclicity of the trace, so $P(UXU^\top) = P(X)$ (and the volume element dX is likewise invariant, orthogonal conjugation being an isometry). Independence of the entries was manifest by construction; invariance fell out of the trace. This is the GOE, sitting in the intersection of the two circles—and the same computation runs in reverse: demanding *both* properties forces the exponent to be built from $\operatorname{tr} X$ and $\operatorname{tr} X^2$ alone, i.e. forces Gaussian entries. At general N nothing changes: $\operatorname{tr} X^2 = \sum_i x_{ii}^2 + 2\sum_{i<j} x_{ij}^2$, so Eq. (1) still factorizes entry by entry (independence) while depending only on a trace (invariance).

Why the *orthogonal* group specifically? Because a real symmetric matrix is diagonalized by an orthogonal matrix (the spectral theorem): the invariance group of the ensemble matches the group that diagonalizes its members. The consequence is worth internalizing—in a rotationally invariant ensemble the eigenvectors carry no information at all, and *everything* statistical lives in the eigenvalues. The family has two siblings, indexed by Dyson's β: complex Hermitian matrices with independent complex-Gaussian entries, invariant under *unitary* conjugation $P(UXU^\dagger) = P(X)$, form the GUE ($\beta = 2$, the relevant ensemble when time-reversal symmetry is broken); quaternionic self-dual matrices with symplectic invariance form the GSE ($\beta = 4$). The GOE ($\beta = 1$) is the one that matches real, time-reversal-symmetric problems—and real-valued data matrices, which is why it is the right reference point for what follows. (The GOE also has a celebrated global law of its own, Wigner's semicircle [1]; we will not need it, because the global law relevant to covariance matrices is Marchenko–Pastur's, derived in Section 5.)

3.2 The local answer: Wigner's surmise

The subtler question is local: take two *adjacent* eigenvalues, λ_i and λ_{i+1}, and let s be their spacing measured in units of the mean spacing. What is the distribution of s? If eigenvalues ignored one another—if they behaved like independent uniform points, a Poisson process—the answer would be

$$P_{\text{Poisson}}(s) = e^{-s}, \tag{2}$$

which is *largest* at $s = 0$: independent points love to clump. Wigner's surmise is that GOE eigenvalues do the opposite:

$$P_{\mathrm{W}}(s) = \frac{\pi s}{2}\, e^{-\pi s^2/4}, \tag{3}$$

which *vanishes* at $s = 0$. Eigenvalues of a random symmetric matrix repel: finding two of them nearly degenerate is not merely rare but suppressed linearly in their distance. This *level repulsion* is the sharpest fingerprint of random-matrix behaviour, and it is the second test we will apply to financial data.

The name "surmise" has a story worth telling, because the calculation behind it is the most instructive small proof in the subject. Asked at a 1956 conference what the spacing distribution of a large random matrix should be, Wigner guessed that the answer for $N \to \infty$ would be close to the answer for the smallest nontrivial case, $N = 2$—which can be computed exactly. Return to the 2×2 GOE matrix of Section 3.1, with its density Eq. (1). Its two eigenvalues are

$$\lambda_{2,1} = \frac{x_1 + x_2}{2} \pm \frac{1}{2}\sqrt{(x_1 - x_2)^2 + 4x_3^2},$$

so the spacing $s = \lambda_2 - \lambda_1$ is

$$s = \sqrt{(x_1 - x_2)^2 + 4x_3^2},$$

and its distribution is the density of Section 3.1 integrated over the surface where the spacing equals s:

$$P(s) = \iiint \frac{dx_1}{\sqrt{2\pi}}\,\frac{dx_2}{\sqrt{2\pi}}\,\frac{dx_3}{\sqrt{\pi}}\; e^{-\frac{1}{2}\left(x_1^2+x_2^2+2x_3^2\right)}\;\delta\!\left(s - \sqrt{(x_1 - x_2)^2 + 4x_3^2}\right). \tag{4}$$

The square root begs for polar-type coordinates. Change variables from (x_1, x_2, x_3) to (r, θ, ψ):

$$x_1 - x_2 = r\cos\theta, \qquad 2x_3 = r\sin\theta, \qquad x_1 + x_2 = \psi,$$

that is,

$$x_1 = \frac{\psi + r\cos\theta}{2}, \qquad x_2 = \frac{\psi - r\cos\theta}{2}, \qquad x_3 = \frac{r}{2}\sin\theta,$$

under which the spacing becomes simply $s = \sqrt{r^2\cos^2\theta + r^2\sin^2\theta} = r$. The change of measure is the 3×3 Jacobian

$$J = \begin{pmatrix} \dfrac{\partial x_1}{\partial r} & \dfrac{\partial x_1}{\partial \theta} & \dfrac{\partial x_1}{\partial \psi} \\ \dfrac{\partial x_2}{\partial r} & \dfrac{\partial x_2}{\partial \theta} & \dfrac{\partial x_2}{\partial \psi} \\ \dfrac{\partial x_3}{\partial r} & \dfrac{\partial x_3}{\partial \theta} & \dfrac{\partial x_3}{\partial \psi} \end{pmatrix} = \begin{pmatrix} \dfrac{\cos\theta}{2} & -\dfrac{r\sin\theta}{2} & \dfrac{1}{2} \\ -\dfrac{\cos\theta}{2} & \dfrac{r\sin\theta}{2} & \dfrac{1}{2} \\ \dfrac{\sin\theta}{2} & \dfrac{r\cos\theta}{2} & 0 \end{pmatrix},$$

whose determinant (expand along the third column, and watch the two cofactors conspire) is

$$\det J = -\frac{r}{4}, \qquad \text{so} \qquad dx_1\,dx_2\,dx_3 = \frac{r}{4}\,dr\,d\theta\,d\psi.$$

The exponent transforms just as cooperatively:

$$x_1^2 + x_2^2 + 2x_3^2 = \frac{(\psi + r\cos\theta)^2 + (\psi - r\cos\theta)^2}{4} + \frac{r^2\sin^2\theta}{2} = \frac{\psi^2 + r^2}{2},$$

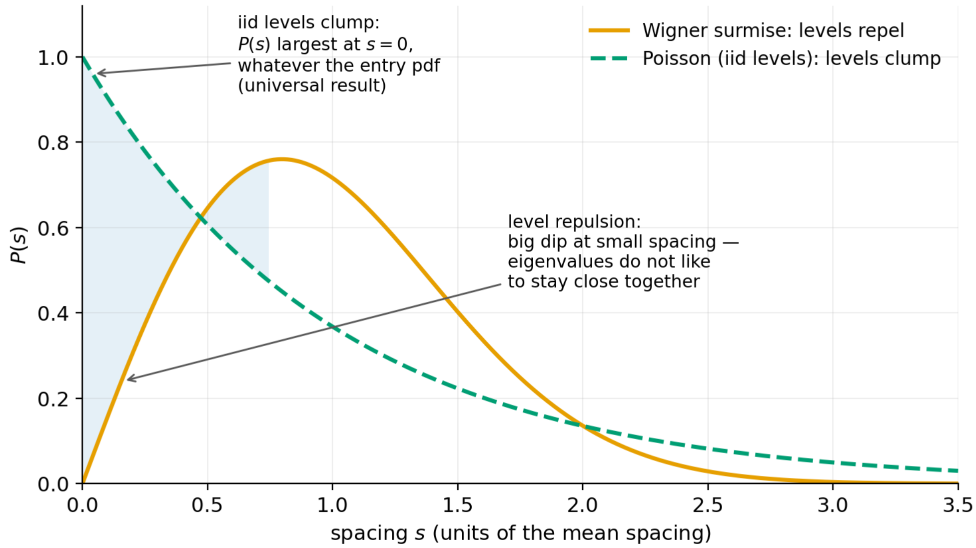

Figure 2: The two competing spacing laws, redrawn from my study notes. Independent (Poisson) levels clump: whatever the pdf of the individual levels, the probability of finding two of them very close together is *largest*—a universal result. Wigner–Dyson levels repel: the surmise vanishes linearly at $s = 0$, the "big dip" at small spacing. Poisson is the statistics of things that ignore each other; Wigner–Dyson is the statistics of things that repel. Which of the two describes the eigenvalues of a financial correlation matrix is an empirical question—answered in Section 6.

with all θ-dependence cancelling. Equation (4) now factorizes completely: the delta function sets $r = s$, the angle integrates to 2π, and the centre-of-mass variable ψ integrates to a Gaussian constant $\int e^{-\psi^2/4} d\psi = 2\sqrt{\pi}$:

$$P(s) = \frac{1}{2\pi\sqrt{\pi}} \cdot \frac{s}{4}\, e^{-s^2/4} \cdot 2\pi \cdot 2\sqrt{\pi} = \frac{s}{2}\, e^{-s^2/4}.$$

This is the exact spacing density of the 2×2 Gaussian matrix. The factor s—the level repulsion—came from the Jacobian: it is the polar area element, and its origin is geometric. For the spacing to vanish, *two* independent quantities ($x_1 - x_2$ and $2x_3$, the two Cartesian components of r) must vanish simultaneously; degeneracy is a codimension-two event, and the area available at radius r shrinks linearly as $r \to 0$. Eigenvalues repel because degeneracies are corners in a space where randomness has room to turn. Finally, the mean of $\frac{s}{2}e^{-s^2/4}$ is $\sqrt{\pi}$; measuring the spacing in units of its mean ($s \to s/\sqrt{\pi}$) turns the result into Eq. (3). Wigner's guess—that $N = 2$ already captures $N = \infty$—turned out accurate to about one percent, which is why a one-page calculation still carries the name of a theorem.

Figure 2 draws the two laws side by side. Everything that matters is at the origin: independent levels pile up there, repelling levels dig a hole. When we meet real data in Section 6, this contrast is what makes the spacing histogram such a sharp diagnostic—no fitted parameter can move a pile-up into a hole.

4 From symmetric matrices to Wishart matrices

Finance hands us a random matrix of a different shape. Collect daily returns of N stocks over T days into $X \in \mathbb{R}^{T \times N}$ (row t = the market on day t), standardize each column, and form the sample correlation matrix

$$S = \frac{1}{T} X^\top X. \tag{5}$$

This matrix is symmetric, but it is emphatically *not* a GOE matrix: its entries are strongly dependent (all built from the same X), it is positive semidefinite by construction, and its natural null model is not "ignorant Hamiltonian" but "returns with no true correlations at all"—X filled with i.i.d. noise of variance σ^2. When the noise is Gaussian, TS is the *Wishart ensemble*, introduced by John Wishart in 1928 [3]—three decades before Wigner—as the sampling distribution of covariance estimates. The eigenvalue theory of Wishart matrices is exactly what a portfolio builder needs: it describes what the spectrum of a sample covariance matrix looks like when there is *nothing* to find, so that anything found beyond it is real.

Why should a portfolio builder care about eigenvalues at all? Because the minimum-variance portfolio

$$w = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}} \tag{6}$$

where,

- Σ is the *covariance matrix* of the stock returns: the $N \times N$ table whose diagonal entry Σ_{ii} is stock i's variance (how much it moves on its own), and whose off-diagonal entry Σ_{ij} is the covariance between stocks i and j (do they move together, or opposite?). It is the same matrix this paper is about throughout—raw or cleaned, whichever version we feed in.
- $\mathbf{1}$ is the column vector of N ones, $(1, 1, \ldots, 1)^\top$: a bookkeeping device whose job is to add things up. For instance, $\mathbf{1}^\top w = w_1 + w_2 + \cdots + w_N$ is the sum of all portfolio weights.
- w is the output: the vector of *portfolio weights*, with w_i the fraction of capital placed in stock i. A negative w_i means the stock is sold short.

The numerator $\Sigma^{-1}\mathbf{1}$ sets the *shape* of the portfolio; the denominator $\mathbf{1}^\top\Sigma^{-1}\mathbf{1}$ is just a number, and dividing by it rescales the weights so they sum to exactly one—the fully-invested condition.

Some intuition for what $\Sigma^{-1}\mathbf{1}$ is doing. If all stocks were uncorrelated, Σ would be diagonal and Eq. (6) would reduce to $w_i \propto 1/\sigma_i^2$: put more money in quieter stocks, less in jumpy ones—perfectly sensible. With correlations present, the inverse also exploits *hedging*: if two stocks move together, holding one long and the other short cancels much of their risk, so Σ^{-1} happily assigns opposite-sign weights to correlated pairs.

And this is exactly why the noise question matters. Inverting Σ flips every eigenvalue: λ becomes $1/\lambda$. The directions the formula trusts most are therefore the ones with the *smallest* estimated eigenvalues—the quietest-looking combinations of stocks. If those small eigenvalues are real, all is well. If they are sampling noise from the Marchenko–Pastur band, the formula piles enormous long–short bets onto mirages that evaporate out of sample—which is precisely the failure mode our backtest exhibits for the raw matrix, and precisely what cleaning the band prevents: it removes the fake "quiet directions" before the inverse can chase them.

This failure mode has a name in the finance literature—*error maximization* [10]—and it is the reason naive $1/N$ weighting is embarrassingly hard to beat [11]. The question "which eigenvalues of S are meaningful?" is therefore not aesthetic; it decides where the money goes. The answer, for Wishart matrices, is the Marchenko–Pastur theorem.

Before stating it, here is what the theorem claims, in everyday terms. Imagine 500 students each answering 100 True/False questions by flipping a coin. There is nothing real to find in these answer sheets: no student knows anything, no two students are copying, no two questions are related. And yet, if you go looking for patterns, you will find them anyway. Purely by luck, some pairs of students will agree suspiciously often; some questions will seem to "move together." Limited data *always* contains accidental patterns. Now, the eigenvalues of a correlation matrix are, roughly speaking, a ranked list of the strongest patterns in the data—the biggest eigenvalue measures the strength of the most prominent pattern, the next one the second most prominent, and so on. Even for pure coin flips, this list will not read "nothing, nothing, nothing": it will contain entries of various sizes, spread over a whole range, each one an accident masquerading as a pattern. What the Marchenko–Pastur theorem supplies is the *exact* description of that range: how strong the strongest accidental pattern can possibly be, how weak the weakest, and how the accidents pile up in between—when there is nothing real to find. It is the fingerprint of pure luck. And once you know precisely what luck looks like, anything in real data that exceeds the fingerprint must be something else.

Theorem 1 (Marchenko–Pastur, 1967 [4])**.** *Let $X \in \mathbb{R}^{T\times N}$ have i.i.d. entries with mean 0, variance σ^2, finite fourth moment, and let $N, T \to \infty$ with $N/T \to \gamma \in (0,\infty)$. Then the eigenvalue histogram of $S = X^\top X/T$ converges to the deterministic density*

$$f(\lambda) = \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{2\pi\gamma\sigma^2\lambda}, \qquad \lambda_\pm = \sigma^2(1 \pm \sqrt{\gamma})^2, \tag{7}$$

supported on $[\lambda_-, \lambda_+]$, plus an atom of mass $1 - 1/\gamma$ at zero when $\gamma > 1$.

This is the fundamental global law of the Wishart world: a deterministic, universal spectral shape, supported on a band whose *location and width depend on the aspect ratio $\gamma = N/T$*. A pure-noise covariance matrix has no eigenvalues outside $[\lambda_-, \lambda_+]$ (up to edge fluctuations vanishing like $N^{-2/3}$); an eigenvalue above λ_+ in real data is structure.

Reading the theorem: signal versus noise. The theorem converts the raw list of eigenvalues into an interpretable scale. For our data, where $\gamma = 0.443$, it predicts that a covariance matrix built from purely uncorrelated returns would still produce eigenvalues spread between 0.11 and 2.78, simply as an artifact of estimating a matrix from a finite sample. Every eigenvalue we observe is then read against this band.

Inside the band. One of our eigenvalues is $\lambda = 0.85$. It sits comfortably inside $[0.11, 2.78]$, meaning a market with no real correlation structure at all would routinely produce an eigenvalue of exactly this size. The direction it points to—some particular combination of stocks—carries no information: it is indistinguishable from what estimation noise alone generates, and should be discarded rather than interpreted.

Above the upper edge. Our second-largest eigenvalue is $\lambda_2 = 11.4$—four times the maximum the noise band allows. No amount of estimation error can produce a value this large; the eigenvalue must correspond to a real, persistent correlation. Examining its eigenvector confirms this directly: it loads heavily on a specific group of related companies that rise and fall together, consistent with a genuine sector or business-group effect. Our largest eigenvalue, $\lambda_1 = 130.2$, is forty-seven times the edge and dominates the entire spectrum. This is the market mode: on any given day, all 548 stocks tend to move in the same direction, and this single eigenvalue is that collective, market-wide co-movement expressed as a number.

Below the lower edge. Eigenvalues can also fall short of the predicted noise floor, and this requires a different, more careful reading than the upper tail. In our data, 41 eigenvalues fall below $\lambda_- = 0.11$, the smallest at 0.043. Two distinct mechanisms account for this:

- **A normalization effect, not a genuine anomaly.** The unadjusted Marchenko–Pastur bound assumes the total variance in the system is distributed without any eigenvalue taking a disproportionate share. In our data this assumption is violated: the two largest eigenvalues, $\lambda_1 = 130.2$ and $\lambda_2 = 11.4$, together account for roughly a third of the total variance in the spectrum. Because total variance is fixed, this concentration at the top necessarily leaves less variance available to the remaining eigenvalues, compressing the entire noise bulk below where the unadjusted formula places it. This is a bookkeeping consequence of the large eigenvalues' dominance, not evidence of missing structure. Rescaling the noise-band prediction to account for the variance already absorbed by λ_1 and λ_2 (Section **??**) removes most of the apparent deficit, confirming that these eigenvalues were never truly anomalous—only measured against an inappropriate baseline.
- **Genuine near-duplication among securities.** Some pairs of securities in our universe are near-duplicates of one another—dual listings, or firms from the same business group—so that a combination of the two barely moves under any market condition. Such a pair produces a real, structurally near-zero eigenvalue: not because of insufficient variance budget elsewhere, but because that specific direction in the data genuinely carries almost no independent information.

The practical rule for eigenvalues below the band therefore differs from the rule above it: rather than treating a low eigenvalue as automatically meaningful, we first correct for the variance already claimed by the dominant eigenvalues, and only interpret whatever residual deficit remains as evidence of genuine near-duplication among specific securities.

Taken together, the reading rule is simple: inside the band, an eigenvalue reflects only estimation noise and carries no structural information; above it, the eigenvalue reflects real, persistent correlation and its eigenvector identifies which assets are involved; below it, the eigenvalue requires a closer look before being classified either way. This single test—whether an eigenvalue exceeds what pure estimation noise could produce—is why the theorem is used far beyond finance, wherever quantities are estimated from limited data, from denoising principal components to genomics. The next section derives Eq. (7) in full.

5 The Marchenko–Pastur law, derived

I learned this derivation as four separate tools that click together, and I present it the same way: each tool first, with its own short proof or proof sketch, then the assembly. The style is a physicist's derivation—every identity exact, the two concentration steps taken on faith with references to where they are made rigorous [5].

5.1 Why the limit holds N/T fixed

Before the tools, the limit itself, because the way it is taken is not a technicality—it is the first real lesson of the subject.

Classical statistics fixes N and sends $T \to \infty$: the law of large numbers then works entry by entry, $S \to \sigma^2 I$, and every eigenvalue converges to σ^2. The spectrum collapses to a point;

estimation is solved; nothing interesting remains. But this limit answers the wrong question for a dataset with $N = 548$ and $T = 1236$. There, each matrix entry carries an error of order $1/\sqrt{T}$—individually harmless—but the matrix has $N(N+1)/2 \approx 150{,}000$ entries, and the eigenvalue spectrum feels all of them at once. Per-entry consistency says nothing about whether the *spectrum* is close to the truth.

The honest limit lets the theory grow with the problem: send *both* $N \to \infty$ and $T \to \infty$ while holding fixed

$$\gamma = \frac{N}{T}, \tag{8}$$

the number of variables per observation—in effect, the number of parameters the data must estimate per datum it supplies. Three observations explain why this is the right scheme.

First, *the ratio is what the answer depends on.* The derivation below produces a limiting density that depends on N and T only through γ: problems with the same aspect ratio share the same limit. Fixing γ is thus like the thermodynamic limit in physics, where particle number and volume diverge together at fixed density; γ is the density, indexing a one-parameter family of genuinely different limits.

Second, *each extreme is degenerate.* As $\gamma \to 0$ the band $[\sigma^2(1-\sqrt{\gamma})^2, \sigma^2(1+\sqrt{\gamma})^2]$ contracts to the point σ^2: the classical limit is recovered as a special case, so Marchenko–Pastur *contains* textbook consistency. As $\gamma \to \infty$ the matrix is overwhelmed by exact zeros (its rank is at most T). All structure lives strictly between, at fixed γ.

Third, *both dimensions must diverge for the answer to be deterministic.* At finite N the eigenvalue histogram is random—each draw of X gives a different one. As N grows, the histogram is an average over ever more eigenvalues and self-averages to a deterministic curve. This is why a theorem about an infinite limit says something about the single market we actually observe: at N in the hundreds, one draw already sits close to the limit curve.

(A note on conventions: some expository accounts use $q = T/N$, the reciprocal. Everything maps across by $\gamma \mapsto 1/q$; our data has $\gamma = 548/1236 = 0.443$.)

5.2 Tool 1: the Stieltjes transform and its inversion formula

The eigenvalue histogram is awkward to manipulate directly; the subject's workhorse encoding is the *Stieltjes transform*. For a probability measure μ on $\mathbb{R}$,

$$m(z) = \int \frac{d\mu(\lambda)}{\lambda - z}, \qquad z \in \mathbb{C}^+, \tag{9}$$

which for the eigenvalues of S is a normalized trace of the *resolvent* $G(z) = (S - zI)^{-1}$:

$$m_N(z) = \frac{1}{N} \operatorname{tr} G(z) = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{\lambda_i - z}. \tag{10}$$

The transform is invertible: the density is recovered by the *Stieltjes inversion formula*

$$f(\lambda) = \frac{1}{\pi} \lim_{\eta \downarrow 0} \operatorname{Im}\, m(\lambda + i\eta). \tag{11}$$

The proof is a picture. Write $z = \lambda + i\eta$ and take imaginary parts inside the integral:

$$\frac{1}{\pi} \operatorname{Im}\, m(\lambda + i\eta) = \int \underbrace{\frac{1}{\pi} \frac{\eta}{(\lambda' - \lambda)^2 + \eta^2}}_{\text{Poisson kernel}} \, d\mu(\lambda'),$$

the convolution of μ with a unit-mass bump of width η. The transform evaluated just above the real axis is therefore a *smoothed histogram* of the eigenvalues, with smoothing scale η; letting $\eta \downarrow 0$ sharpens the smoothing away and returns the density. The plan of the whole derivation is now visible: compute $m(z)$ by linear algebra, then push z down onto the real axis.

5.3 Tool 2: the Sherman–Morrison rank-one update

For invertible A and vectors u, v with $1 + v^\top A^{-1} u \neq 0$,

$$(A + uv^\top)^{-1} = A^{-1} - \frac{A^{-1} uv^\top A^{-1}}{1 + v^\top A^{-1} u}, \tag{12}$$

verified by multiplying both sides by $A + uv^\top$. The corollary that does all the work here comes from taking $v = u$ and sandwiching with $u^\top \cdot u$:

$$u^\top (A + uu^\top)^{-1} u = \frac{u^\top A^{-1} u}{1 + u^\top A^{-1} u}. \tag{13}$$

The relevance: a sample covariance matrix is a *sum of rank-one pieces*, one per trading day, $S = \frac{1}{T} \sum_t x_t x_t^\top$. Sherman–Morrison is the scalpel that removes a single day from the matrix and expresses exactly what that day contributed.

5.4 Tool 3: Sylvester's determinant identity

For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times m}$,

$$\det(I_m + AB) = \det(I_n + BA). \tag{14}$$

Applied to $S = X^\top X/T$ and its $T \times T$ companion $XX^\top/T$, it gives

$$\det\Big(\tfrac{1}{T} X^\top X - zI_N\Big) = (-z)^{N-T} \det\Big(\tfrac{1}{T} XX^\top - zI_T\Big) :$$

the "assets $\times$ assets" and "days $\times$ days" matrices share all nonzero eigenvalues, the larger being padded with exact zeros. Two consequences matter. When $\gamma > 1$ (more assets than days), S has at least $N - T$ zero eigenvalues—the atom of mass $1 - 1/\gamma$ in Theorem 1, delivered by pure linear algebra before any limit is taken, and precisely the singularity that makes Eq. (6) undefined for a one-year estimation window ($T = 252 < N = 548$). And the identity explains why the two ratio conventions (N/T versus T/N) coexist in the literature: the two spectra determine each other.

5.5 Tool 4: concentration of quadratic forms

If $x \in \mathbb{R}^N$ has i.i.d. entries of mean 0, variance σ^2, independent of a symmetric M with bounded norm, then

$$\frac{1}{T} x^\top M x = \frac{\sigma^2}{T} \operatorname{tr} M + O_P\Big(\tfrac{\sqrt{N}}{T}\Big). \tag{15}$$

The mean is a one-line computation ($\mathbb{E}\, x^\top M x = \sigma^2 \operatorname{tr} M$; cross terms vanish); the fluctuation bound is a second-moment (or Hanson–Wright) estimate. Two points deserve emphasis, because they are where probability actually enters the derivation. The independence of x from M is essential—and is exactly what the leave-one-out construction of the next subsection arranges. And the error vanishes only because *both* dimensions diverge: this is Section 5.1 doing work inside the proof, not just in the statement of the theorem.

5.6 The assembly

Write S as a sum over days and remove day t:

$$S = \frac{1}{T}\sum_{t=1}^{T} x_t x_t^\top, \qquad S_{(t)} = S - \tfrac{1}{T}x_t x_t^\top, \qquad G_{(t)} = (S_{(t)} - zI)^{-1},$$

noting that x_t is independent of $S_{(t)}$.

Step 1 (Sherman–Morrison). Eq. (13) with $A = S_{(t)} - zI$, $u = x_t/\sqrt{T}$:

$$\frac{1}{T}x_t^\top G\, x_t = \frac{\frac{1}{T}x_t^\top G_{(t)} x_t}{1 + \frac{1}{T}x_t^\top G_{(t)} x_t}. \tag{16}$$

Step 2 (concentration). Because x_t is independent of $G_{(t)}$, Tool 4 gives

$$\frac{1}{T}x_t^\top G_{(t)} x_t \approx \frac{\sigma^2}{T}\operatorname{tr} G_{(t)} \approx \sigma^2 \frac{N}{T}\, m_N(z) = \sigma^2\gamma\, m_N(z), \tag{17}$$

the second approximation because a rank-one change moves the trace of a resolvent by $O(1)$, negligible after division by T. Every day contributes the *same* macroscopic quantity: this is self-averaging in action.

Step 3 (a trace identity closes the loop). Compute $\operatorname{tr}(SG)$ twice. Algebraically,

$$\operatorname{tr}(SG) = \operatorname{tr}[(S - zI)G] + z\operatorname{tr} G = N + zN\, m_N(z).$$

Day by day, using Steps 1–2,

$$\operatorname{tr}(SG) = \sum_{t=1}^{T} \frac{1}{T}x_t^\top G x_t \approx T \cdot \frac{\sigma^2\gamma m_N}{1 + \sigma^2\gamma m_N}.$$

Equate, divide by $N = \gamma T$, and pass to the limit m:

$$1 + zm = \frac{\sigma^2 m}{1 + \sigma^2\gamma m}. \tag{18}$$

Step 4 (solve). Eq. (18) is a quadratic,

$$\gamma\sigma^2 z\, m^2 + \big(z - \sigma^2(1-\gamma)\big)m + 1 = 0,$$

with solution

$$m(z) = \frac{\sigma^2(1-\gamma) - z + \sqrt{\big(z - \sigma^2(1-\gamma)\big)^2 - 4\gamma\sigma^2 z}}{2\gamma\sigma^2 z}, \tag{19}$$

the branch fixed by $m(z) \sim -1/z$ at large $|z|$ (all eigenvalues are finite). The discriminant factorizes,

$$\big(z - \sigma^2(1-\gamma)\big)^2 - 4\gamma\sigma^2 z = (z - \lambda_-)(z - \lambda_+), \qquad \lambda_\pm = \sigma^2(1 \pm \sqrt{\gamma})^2,$$

(check the sum and product), and this factorization is the birthplace of the band edges.

Step 5 (Stieltjes inversion). For real λ outside $[\lambda_-, \lambda_+]$ the discriminant is positive, $m(\lambda + i0)$ is real, and Eq. (11) gives zero density. Inside the band the square root turns imaginary, $\sqrt{\cdot} = i\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}$, and

$$f(\lambda) = \frac{1}{\pi} \operatorname{Im} m(\lambda + i0) = \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{2\pi\gamma\sigma^2\lambda}, \tag{20}$$

which is Theorem 1. For $\gamma > 1$, Eq. (19) has a simple pole at $z = 0$ with residue $1 - 1/\gamma$: the atom of zeros, now appearing analytically and agreeing exactly with the count Sylvester's identity gave by linear algebra—a satisfying consistency check between two of the tools. Note finally that Gaussianity was never used: only two moments entered (through Tool 4), so the law is universal, and in particular applies to fat-tailed financial returns.

6 Using it: NSE equities

Theory in hand, I turn to the data I assembled to test it. The claim under examination is concrete: *the eigenvalues of the sample correlation matrix that fall inside the Marchenko–Pastur band are genuinely noise*. The evidence comes in three layers of increasing strictness. The *global* layer checks that the band's eigenvalues are distributed the way the theorem says noise is distributed. The *local* layer checks that they interact the way Wigner says noise eigenvalues interact. Both are consistency checks—necessary, but a sufficiently clever skeptic could still insist that some faint, usable structure hides inside the band. So the third, *economic* layer makes the claim falsifiable: build real portfolios that discard the band's contents and portfolios that trust them, and let unseen future data decide. If the band carried information, discarding it would hurt; if it is noise, discarding it should help. This section runs all three tests.

6.1 The data

Daily log returns of the constituents of the Nifty Total Market index (Nifty 500 + Microcap 250), 2 July 2021 to 30 June 2026, from Yahoo Finance with split/dividend adjustment. Cleaning removed stocks that listed mid-sample (187, the 2021–24 IPO wave), a few stale or gap-ridden series, and thirteen stocks whose price series carry verified vendor errors (bonus/split adjustments stamped on the wrong date; unadjusted demerger cliffs)—errors that would have injected fictitious daily moves as large as -75% into every covariance estimate. Extreme returns were verified against news records and never clipped; the largest survivors (the Adani episode of February 2023, the election-day crash of June 2024) are real. The result: $N = 548$ stocks, $T = 1236$ days, $\gamma = 0.443$; returns standardized per stock, so S is a correlation matrix and $\sigma^2 = 1$ under the noise null.

6.2 The global test: the eigenvalues sit where noise sits

With $\gamma = 0.443$ the noise band is $[0.11, 2.78]$. Figure 3 shows what five years of Indian equities actually produce: 496 of 548 eigenvalues (90.5%) inside the band; 11 above it, led by $\lambda_1 = 130.2$—the market mode, whose eigenvector weights all 548 stocks with the same sign and which alone carries 23.8% of total variance. Comparing the bulk's *shape* to Eq. (7) requires one refinement from the applied literature [6]: the signal modes absorb a third of the trace, so the noise bulk lives on the remaining two-thirds and its effective variance is below one. Fitting σ^2 gives $\hat{\sigma}^2 = 0.560$, and Figure 4 shows the result: the MP density tracks the empirical bulk closely.

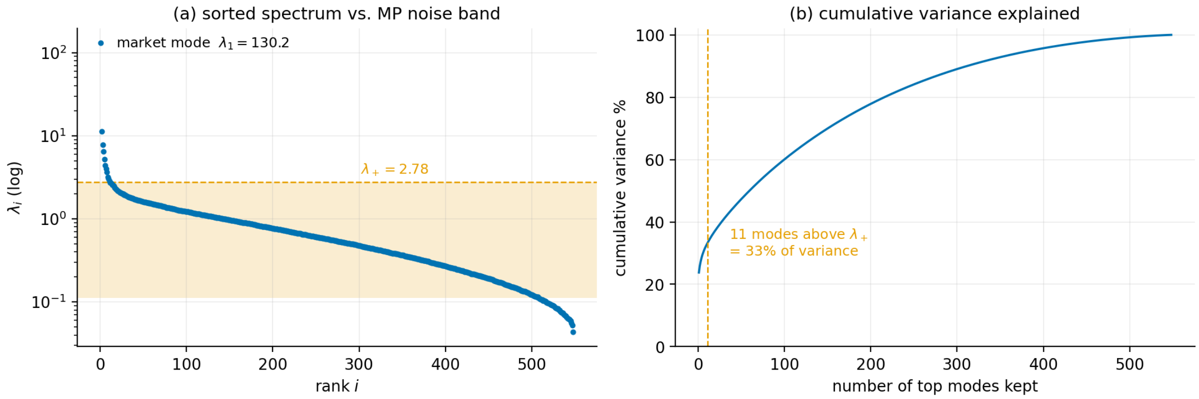

Figure 3: The spectrum meets the theorem. (a) Sorted eigenvalues of the NSE correlation matrix against the $\sigma^2 = 1$ Marchenko–Pastur band (shaded): the bulk hugs the band; 11 eigenvalues escape above, led by the market mode $\lambda_1 = 130$. (b) Those few escapees carry a third of all variance.

6.3 The local test: the eigenvalues repel the way noise repels

The density is a global statement, and Section 3 taught that random matrices make a second, sharper promise: level repulsion. So I tested it. Take the 486 eigenvalues inside the fitted band, unfold them (rescale so the local mean spacing is one, using the fitted MP law as the density), and histogram the 485 adjacent spacings. Figure 5 shows the result against the two candidates of Section 3.2: the data follow Wigner's surmise—the same curve derived from the 2×2 matrix—and reject the Poisson alternative decisively (Kolmogorov–Smirnov distance 0.027 against the surmise, versus 0.209 against Poisson; compared with noise simulations of matching size, the surmise is fully consistent while Poisson is excluded at eleven standard deviations). The unfolding-free ratio statistic $\tilde{r}_i = \min(s_i, s_{i+1}) / \max(s_i, s_{i+1})$ [9] agrees: $\langle \tilde{r} \rangle = 0.536$, against 0.531 for GOE and 0.386 for Poisson. The noise band does not merely *sit* where the Marchenko–Pastur theorem says noise sits; its eigenvalues *repel* exactly as Wigner said noise eigenvalues repel. Both layers of the theory hold in this market, and together they license what comes next.

6.4 The economic test: does the noise band price like noise?

The two spectral tests certify that the band *looks* like noise. They cannot certify that it is *useless*: a curve can fit beautifully and still hide a sliver of real, exploitable information. So the final test leaves curve-fitting behind and stakes money— simulated money, but scored with complete honesty—on the answer.

Here is the experiment in plain terms. We do something deliberately drastic to the covariance matrix: keep the handful of eigenvalues that stick out above the noise band (about eleven of 548, in a typical year of data), and *erase every distinction* the data drew among the hundreds of directions inside it, replacing them all by one flat average level—chosen so that total variance is preserved: we reattribute risk, we never delete it. Now ask what this act of vandalism does to real portfolios. If the band contains genuine information about risk, destroying it must make our portfolios *worse*. If the band is noise, then the "information" we erased was never information at all—it was the statistical equivalent of static—and erasing it should actually *help*, because the portfolio optimizer will no longer be fooled by it. Both outcomes are possible in advance. The future decides. (One case is settled before the race begins: with one year of data—252

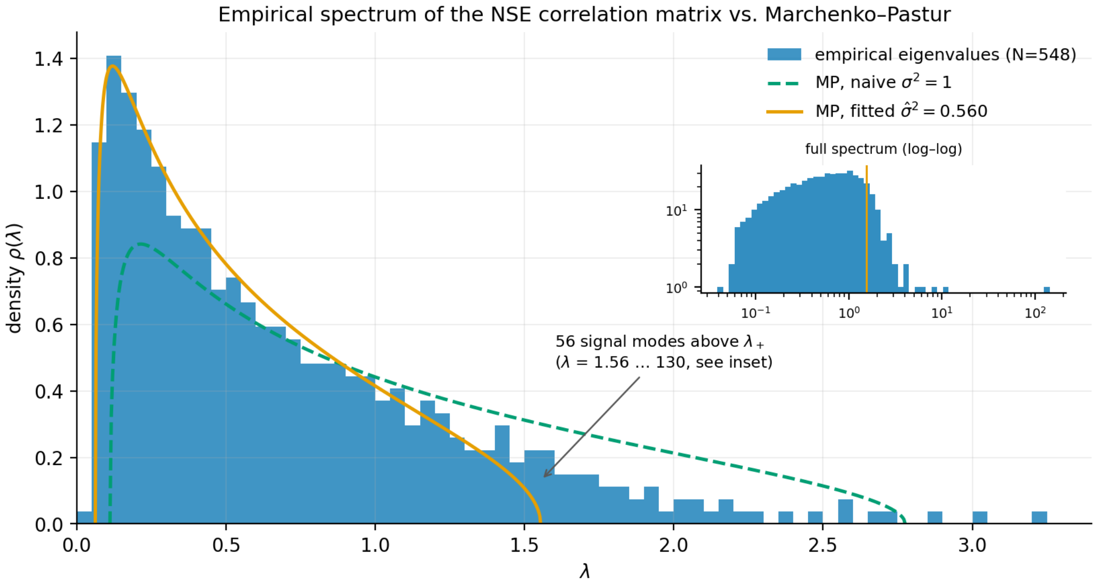

Figure 4: The global test: eigenvalue histogram against the Marchenko–Pastur density, Eq. (7). The naive $\sigma^2 = 1$ curve (dashed) is too wide because the signal modes absorb a third of the trace; with the effective bulk variance fitted ($\hat{\sigma}^2 = 0.560$, solid) the law tracks the data closely (Kolmogorov–Smirnov distance 0.024). Inset: full spectrum, log–log.

days—and 548 stocks, the raw covariance matrix breaks down completely, like trying to pin down 548 unknowns from 252 equations. The cleaned matrix has no such problem: filling in the noise floor is exactly what repairs it.)

The test is run as a four-year contest between four imaginary fund managers. Each has the same mandate—hold the least-risky fully invested portfolio you can build, Eq. (6)—and each sees only the past, never the future. Once a month, every manager re-estimates risk from the trailing window of returns and rebalances; the portfolio is then held for the next month, whose returns none of them has seen, and its day-by-day performance is recorded. The four differ only in how they treat the covariance matrix. **Manager 1 (raw)** trusts the estimate exactly as it comes (using the standard mathematical workaround, a pseudo-inverse, when it breaks). **Manager 2 (MP-cleaned)** erases the noise band, as above, with the threshold set purely from past data. **Manager 3 (Ledoit–Wolf)** applies the standard statistical remedy [12]: nudge every eigenvalue toward the average—cautious and sensible, but blind, with no theory of which eigenvalues are noise. **Manager 4 (equal weight)** ignores covariances entirely and splits the money evenly across all 548 stocks—the famously hard-to-beat benchmark [11]. Every trade is charged a realistic transaction cost. Because the mandate is low risk, the score is simple: *how bumpy was each manager's ride*—the realized volatility of each portfolio over roughly a thousand days of unseen data. To make sure nothing hinges on an arbitrary choice, the whole contest is run twice, once with a one-year estimation window and once with three years. Table 1 and Figure 6 report both runs.

The contest is not close. Manager 1, who trusted everything the sample said, had the wildest ride: 19.4% annualized volatility—worse than ignoring covariances altogether. The reason is visible in Figure 6b, and it is the error maximization of Section 4 caught on camera. The noisy matrix keeps "discovering" combinations of stocks that look magically safe, and the optimizer

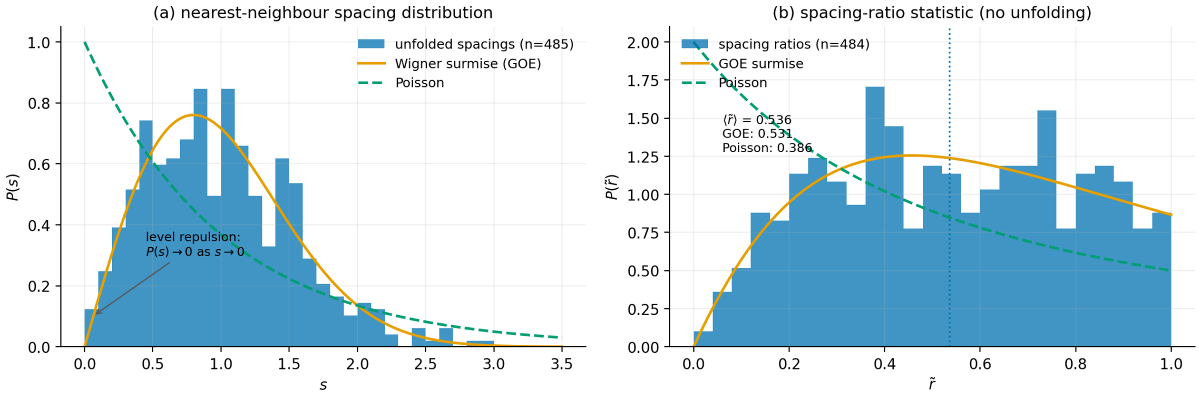

Figure 5: The local test: Wigner's surmise on real data. (a) Distribution of spacings between adjacent noise-band eigenvalues (unfolded to unit mean spacing) against Eq. (3) and the Poisson alternative: the data show exactly the level repulsion of Section 3.2. (b) The spacing-ratio statistic, which requires no unfolding, agrees: $\langle \tilde{r} \rangle = 0.536$ against the GOE value 0.531 and the Poisson value 0.386.

Table 1: Out-of-sample minimum-variance performance, NSE equities. Annualized volatility and return in percent; Sharpe net of 10 bps costs; turnover one-way per rebalance; leverage is average $\sum_i |w_i|$. Top: $T_{\text{est}} = 252$ (47 rebalances, Jul 2022–Jun 2026; raw matrix singular, pseudo-inverted). Bottom: $T_{\text{est}} = 756$ (23 rebalances, Jul 2024–Jun 2026; raw invertible).

	Ann. vol	Ann. ret	Sharpe (net)	Turnover	Leverage
$T_{\text{est}} = 252$					
Raw (pseudo-inverse)	19.37	24.67	0.93	5.36	6.86
MP-cleaned	**7.81**	13.57	1.61	0.84	2.79
Ledoit–Wolf	9.29	16.64	1.57	1.67	3.83
Equal weight	17.66	29.27	1.65	0.07	1.00
$T_{\text{est}} = 756$					
Raw	13.01	−8.21	−0.94	3.50	7.95
MP-cleaned	**8.87**	−4.81	−0.62	0.64	3.24
Ledoit–Wolf	10.23	−3.71	−0.55	1.65	5.28
Equal weight	19.36	6.61	0.34	0.07	1.00

piles into them: for every rupee of capital, Manager 1 holds about seven rupees of offsetting long and short positions on average, spiking to nearly *forty* in early 2026—precisely when the portfolio's realized risk explodes. Those magically safe combinations were mirages of sampling error, and they evaporated in the very next month, month after month, forcing frantic, expensive trading to chase their replacements. Manager 2, who erased the noise band, had the calmest ride of all four: 7.81%—less than half the raw manager's risk—achieved with modest positions and the least trading of any covariance-based strategy. Manager 3's blind shrinkage helped (9.29%) but gave up a meaningful gap to Manager 2 while trading twice as much; Manager 4's equal weights realized 17.7%. And the gaps are far too large to be luck: a standard statistical comparison of day-by-day risk (a Diebold–Mariano test) puts the odds that chance alone explains Manager 2's edge over Managers 3 and 4 below one in ten thousand, and below one in six hundred against Manager 1. The second run of the contest, with three-year windows—where the raw matrix works without any mathematical workaround—produces the same finishing order (8.87% vs.

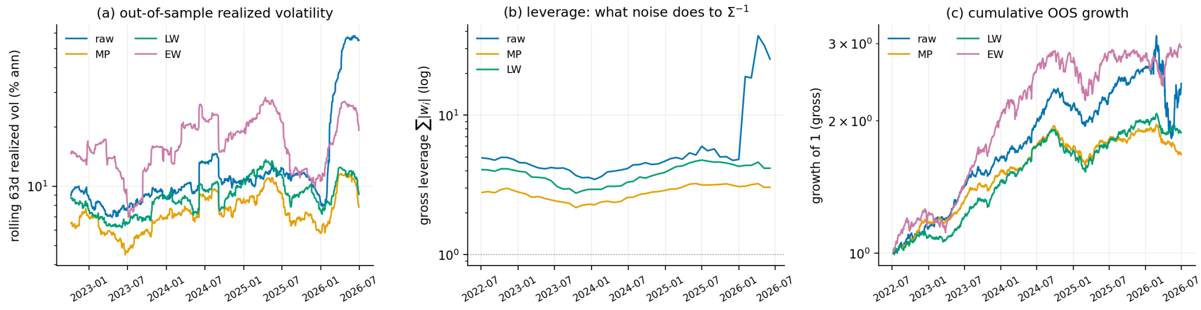

Figure 6: The backtest through time ($T_{\text{est}} = 252$). (a) Rolling realized volatility: the MP-cleaned portfolio is lowest almost throughout. (b) Gross leverage: the raw matrix's noise-driven bets spike to $\sim$40× in early 2026, exactly when its realized risk explodes. (c) Cumulative out-of-sample growth.

10.23% vs. 13.01%), so nothing here is an artifact of the one-year window. Even after trading costs, Manager 2 has the best risk-adjusted performance of the three covariance-based strategies.

Pause on how strange this result would be if the band were not noise. Manager 2 threw away 98% of the distinctions in the estimated matrix—the most destructive thing a manager could do to their own risk model—and *won because of it*. Had the band contained real information, that act would have placed Manager 2 last, not first: the experiment was completely free to falsify the noise hypothesis, and instead it confirmed it. The three layers of evidence now interlock. The band's eigenvalues are *distributed* like noise (Figure 4); they *interact* like noise (Figure 5); and—the strictest test, because it risks being wrong about the future—they *price* like noise: portfolios that discard them beat portfolios that trust them, portfolios that shrink them blindly, and portfolios that ignore covariances altogether. The noise band is legit. (The fine print, stated honestly: the stock list is today's index membership, so the sample tilts toward companies that survived—a bias that affects all four managers identically; the portfolios are unconstrained and long–short, the setting where the quality of a covariance matrix matters most; trading costs are stylized; and this is one market over one five-year period.)

7 Conclusion: what I learned

Three things, in ascending order of surprise. First, the classical theory is a single connected story: Wigner's move—model ignorance as randomness, then compute what is universal—transfers from nuclear Hamiltonians to sample covariance matrices with only a change of ensemble, from GOE to Wishart, and a change of limiting spectral law, to Marchenko–Pastur; the proofs themselves are assembled from a small kit (a transform that smooths the spectrum, a rank-one update, a determinant identity, a concentration estimate) whose members are each elementary. Second, the limit $N, T \to \infty$ at fixed $\gamma = N/T$ is not an analyst's convenience but the entire content of the modern viewpoint: γ is the price of estimation measured in parameters per datum, the classical theory is the $\gamma \to 0$ edge of the family, and real datasets live well inside it. Third—and this was not obvious to me before doing it—the theory is *quantitatively* true of a real emerging market: five years of NSE returns produce a spectral bulk with the Marchenko–Pastur shape and Wigner's level repulsion, and acting on that diagnosis, by the simplest possible filter, beats both the raw estimate and the standard shrinkage remedy out of sample, by a wide and statistically significant margin. Random matrix theory begins as a confession of ignorance; applied carefully, it ends as an instrument precise enough to trust with money.

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